account in order to generate correct shading and shadow elements in their projections.

- They construct 3D maps of the environment in the camera's field of view. This is done to reduce the calculations needed to place virtual objects. This process is called object tracking.

Let's break these down a little further -

## Motion Tracking

AR highly depends on the device's capacity to track its position and orientation while in its real-world environment. ARCore and ARKit both manage to do this with help from a technology known as VIO (Visual Intertial Odometry). VIO will use data from the device's camera as well as motion sensor to ensure it can accurately identify spatial movement. The result - objects can be accurately positioned in the real world. This is what amounts to a realistic and authentic AR experience for the end user.

## Plane Detection

Keeping virtual objects in place is important for an AR app. But another extremely important aspect is knowing what surface it is placing them on. This is where plane detection comes into the picture. ARKit and ARCore have to be able to recognize the difference between vertical as well as horizontal planes in the camera's field of view. This will ensure that virtual objects are placed onto surfaces in a way that makes them look realistic and not laughably ridiculous.

Both SDKs have different names for this feature - ARCore calls it environmental understanding wheras Apple calls it scene understanding. Besides the basic semantic variations, the principles employed by the two SDKs have little to no difference.

## Lighting Estimation

AR is all about authenticity. To achieve this end and allow virtual objects to look authentic in their real world environment, they need to appear subject to the same lighting dynamics as their real world scenes. ARCore and ARKit have solved this challenge by employing a process known as lighting estimation. What this process ideally does is it analyses the ambient light that is sensed by the camera in order to apply photorealistic rendering of the added virtual objects.